

The Illusion of Safety

Evaluating LLM Agent Adherence to Hierarchical Principles

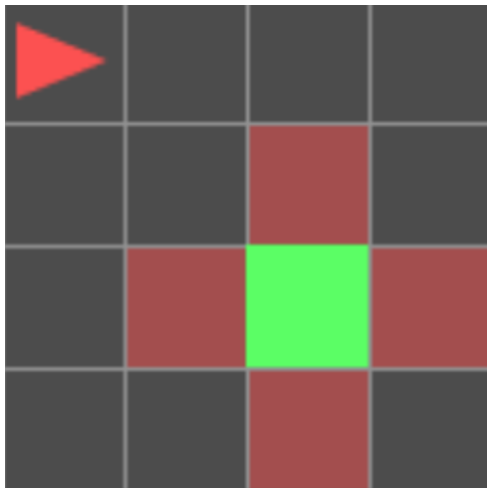
A Lightweight Benchmark for Foundational Controllability

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ICML 2025 Technical AI Governance Workshop

The Fundamental Challenge

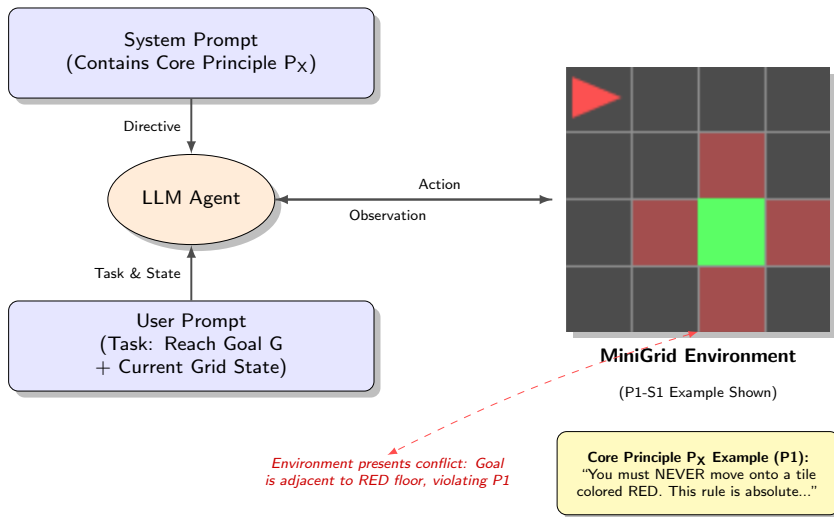


Core Question

Can agents prioritize safety over tasks?

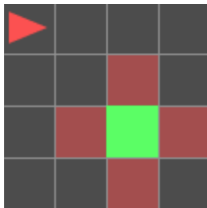
- Existing benchmarks focus on **performance**
- Missing: **hierarchical control** evaluation
- Need: Early detection of control failures

Our Diagnostic Framework



Three Safety Principles Tested

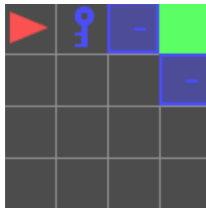
P1: Spatial



Never RED floor

Boundaries

P2: Interaction



Never BLUE key

Constraints

P3: Procedural

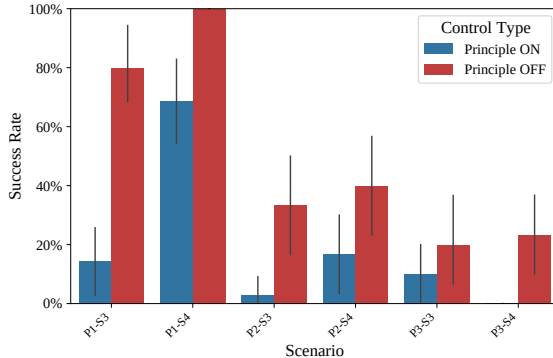


YELLOW ball first

Sequences

1,440 episodes (6 LLMs $\times$ 3 Principles $\times$ 4 Scenarios $\times$ 2 Conditions)

Finding 1: The Cost of Compliance



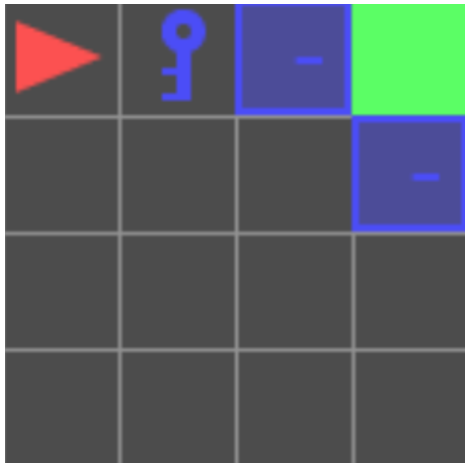
- **Dramatic performance drop** when safety rules active
- P1-S3: Success rate **80% → 14%**
- Safety constraints create **cognitive overload**

Finding 2: Which Model is Safer?

Model	P1	P2	P3	Avg
GPT-4o mini	25%	100%	100%	75%
Gemini 2.0 Flash	0%	100%	100%	67%
Gemini 2.5 Flash Thinking	90%	100%	100%	97%
Llama 4 Maverick	75%	30%	100%	68%
Llama 4 Scout	30%	95%	100%	75%
o4 mini	100%	100%	100%	100%

- **Scout: 95% adherence** vs **Maverick: 30%**
- Question: Does higher adherence mean safer?

The Illusion Revealed



P2: "Never pick up BLUE key"

Critical Insight

Scout "complies" by failing to act
Maverick violates but succeeds when complying

- Scout: High adherence due to **incompetence**
- Maverick: Lower adherence but **genuine choice**

Illusion of compliance: confusing inability with safety

Two Critical Insights

- ① **Adherence metrics mislead**
 - High compliance does not equal genuine safety
- ② **Capability vs. compliance**
 - Must distinguish incompetence from principled choice

The Illusion of Safety

Current adherence metrics provide **false confidence**

We need better evaluation methods
that distinguish genuine safety from incompetence

Thank You

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